

Impact of charging infrastructure on electric vehicle adoption: A synthetic population approach

Lavan T. Burra^{*}, Mohammad B. Al-Khasawneh, Cinzia Cirillo

Department of Civil and Environmental Engineering, University of Maryland, United States

ARTICLE INFO

Keywords:

Electric vehicle
Charging infrastructure
Synthetic population
DC fast charging

ABSTRACT

There is limited availability of travel survey data on households with electric vehicles (EVs) and a lack of evidence on factors influencing EV ownership levels at a finer geographic level, which are crucial for optimizing public charging infrastructure investments. To address this gap, we propose an integrated approach utilizing a discrete choice model and a Bayesian network-generated synthetic population. Applied to Maryland, the model analyzes the impact of public charging stations (level-2 and DC fast chargers) on EV ownership at the census tract level. Access to fast charging, workplace charging, and the possibility of teleworking are key factors influencing EV ownership. The model, applied to the synthetic population, predicts higher EV growth in suburban regions compared to urban areas and a larger increase in EV adoption among high-income groups. This highlights potential disparities in EV adoption and demonstrates the application of this methodology in understanding micro-level EV adoption rates for informing targeted policies and infrastructure development to promote equitable adoption.

1. Introduction

Traditional household travel surveys available regionally or nationally often lack sufficient data on households with electric vehicles (EVs), limiting their utility in understanding EV adoption nuances. Consequently, researchers often rely on stated-preference survey data, which capture hypothetical choices rather than the actual decisions (revealed preferences) of respondents. Additionally, understanding micro-level EV adoption rates is crucial for informing targeted policies and infrastructure development aimed at promoting equitable EV ownership across different economic groups and geographic areas.

The recent allocation of \$7.5 billion in the US as part of the 2021 Infrastructure Investment and Jobs Act (IIJA) for building electric vehicle charging stations along major highways and within low-income and disadvantaged communities, is currently in the initial rounds of funding allocation to states. Given the efforts to expand the public charging network and electrify the light-duty vehicle fleet, understanding the impact of building new public charging stations on EV ownership among different economic groups at a finer geographic level becomes critical.

However, most studies in the literature rely on choice experiments or

stated-preference survey data to investigate the potential role of infrastructure (see [Hardman et al. \(2018\)](#) for an overview of such studies). More recent studies use aggregate car registration data to analyze the effect of financial incentives and access to public charging stations on EV sales (A review of such studies can be found in [Sheldon \(2022\)](#) and [Hardman et al. \(2017\)](#)). Only a few studies utilize revealed preference data that includes socio-demographic information on households (e.g., [Xing et al. \(2021\)](#), [Linn \(2022\)](#), [Jenn et al. \(2020\)](#)). Acquiring such granular datasets can be expensive, and collecting them directly is resource-intensive. Furthermore, little is known about how low-income groups and rural residents respond to the availability of public charging infrastructure (as highlighted in [Hsu and Fingerman \(2021\)](#) and [Canepa et al. \(2019\)](#)).

To investigate this, researchers can use household travel survey data, which typically includes information about the vehicle holdings of households in the sample. However, since the adoption rates of EVs are gaining momentum and the latest travel survey data were collected before 2020 and include a very small share of EVs,¹ estimating reliable transportation models or statistics for policymakers becomes challenging. To overcome this limitation, we propose an integrated synthetic population approach that provides detailed sociodemographic and

^{*} Corresponding author.

E-mail addresses: ltburra@umd.edu (L.T. Burra), mbmk2020@umd.edu (M.B. Al-Khasawneh), ccirillo@umd.edu (C. Cirillo).

¹ In the 2017 National Household Travel Survey, less than 1% of surveyed vehicles are EVs.

socioeconomic information at individual and household levels, preserving the dependency structure between variables of interest and the aggregated statistics of the actual population. This technique has been successfully used in transportation microsimulation systems, such as activity-based models (Barthelemy and Toint, 2015; Wang et al., 2021; Nejad et al., 2021).

Through this, the proposed research aims to address the following questions in the context of EV ownership. First, what factors influence households' decision to own an EV, and how does access to public charging stations influence EV ownership rates? Second, how does increasing the availability of public charging stations influence EV adoption across different income groups and residential locations? These questions are addressed by taking Maryland as the case study.

Maryland's geographical proximity to major metropolitan areas, its households covering a wide range of socioeconomic backgrounds—from affluent urban areas to rural communities with varying income levels—and the unique mix of urban, suburban, and rural regions, present an opportunity to study EV ownership in this diverse landscape. This can offer insights into how income, access to charging infrastructure, and other factors impact EV adoption among different demographic groups. Such information can help inform targeted strategies for EV infrastructure deployment and expedite the EV demand while considering equity considerations.

2. Literature

There is a growing body of literature that has examined the factors important to plug-in electric vehicle (PEV) consumers. Several studies, based on consumer surveys and PEV sales data, have highlighted inadequate access to charging infrastructure and longer battery charging times as significant barriers to large-scale EV adoption, among other reasons (Egbue and Long, 2012; Sheldon et al., 2019; Hardman et al., 2018; Li et al., 2020; Funke et al., 2019; Sierzechula et al., 2014; Lutsey et al., 2016; Slowik and Lutsey, 2017; Bobeth and Matthies, 2018; Hackbarth and Madlener, 2013; Nazari et al., 2018; Globisch et al., 2019; Greene et al., 2020).

Specifically, studies focused on consumer preferences reveal that regular charging options, such as home or workplace charging, play a critical role in PEV ownership during the early phase of adoption (Hardman et al., 2018; Sheldon et al., 2019; Funke et al., 2019; Jenn et al., 2020). Hardman et al. (2018) find that, on average, 50–80% of EV charging occurs at home.² Funke et al. (2019) in a meta-analysis of studies across different countries, finds that home charging alone is sufficient for early EV buyers who have access to parking and can charge their car overnight. Indicating public slow charging for these buyers would act as a substitute for home charging.

Research on the EV market in the US (Li et al., 2017; Narassimhan and Johnson, 2018), Norway (Springel, 2021; Schulz and Rode, 2022), Germany (Sommer and Vance, 2021), and China (Li et al., 2022), using revealed preference or vehicle registration data, has shown a positive effect of charging infrastructure on EV adoption rates. However, most of these studies do not differentiate between different levels of charging, with (Sommer and Vance, 2021) being the exception. It is important to consider nuances between different levels of public charging (mainly level-2 and DC fast charging) to better understand EV adoption rates (Cole et al., 2021). Li et al. (2017) highlights the indirect network effects between EVs and public charging infrastructure using EV registration data aggregated at the Metropolitan statistical areas. Studies based on GPS data/real-world car usage data find that among public charging stations, access to DC fast charging has a greater impact on PEV adoption than level-2 or other slower charging stations (Neaimeh et al., 2017; Levinson and West, 2018; Lee et al., 2019).

In summary, among various factors, access to charging at the home

location is found to be the most important factor for consumers to own an EV, followed by workplace charging, and public charging stations, which is still a key parameter in encouraging EV ownership (Hardman et al., 2018; Sheldon et al., 2019; Nicholas et al., 2017; Plötz and Funke, 2017; Nicholas and Tal, 2017).

Among the aforementioned literature, the majority of studies either utilize stated-preference consumer survey data or aggregated vehicle registration data, which do not provide information on household demographics. The present study, however, utilizes household travel survey data from the state of Maryland to understand factors influencing households' choice of owning an EV. This data enables us to evaluate how their EV ownership changes across different area types and demographic groups with increased accessibility to public charging.

3. Data description

The data required for this study is constructed from different sources that provide information on EV ownership at the household level, public charging station location and characteristics, household sociodemographics, and travel patterns in the state of Maryland, USA.

3.1. Maryland regional travel survey

The primary data source is the Regional Travel Survey (RTS), conducted by the Metropolitan Washington Council of Governments (MWCOC), covering the DMV regions from October 2017 to December 2018. Additionally, the study is complemented by the Maryland Travel Survey (MTS) conducted by the Baltimore Metropolitan Council (BMC), spanning from April 2018 through May 2019. For this study, we focus only on the state of Maryland.

During these surveys, travel diary information was collected from 17,152 households that own 31,186 vehicles, providing detailed insights into household composition, residential locations, vehicle holdings, and their characteristics. Furthermore, the surveys recorded a one-day travel diary of the household residents, which includes information about the start and end time of each trip, trip length, and mode of transport. The data encompasses households from various area types, including transit and low-income areas, making it highly relevant to our study. Moreover, the survey includes valuable information about the possibility of charging EVs at work (if the households own any) and the time spent teleworking.

To ensure data consistency and reliability, we excluded households that do not have any income information. As a result, the final dataset used for analysis consists of 16,757 households after the cleaning process.

Table 1 presents the summary statistics of sociodemographic and location characteristics of the household sample in the survey data. On average, each household owns approximately two cars. Around 94% of

Table 1
Summary statistics of households and location attributes.

Variable	N.of valid Obs	Mean	Std. Dev.	Min	Max
Household size	16,757	2.19	1.18	1	8
Number of workers	16,757	1.17	0.90	0	7
Household income	16,757	5.67	1.99	1	8
Number of vehicles	16,757	1.85	1.08	0	12
Average daily trips per person	16,757	3.44	2.14	0	41
Number of daily transit trips	16,757	0.76	1.82	0	28
Home office dummy	16,757	0.32	0.47	0	1
Home ownership	16,757	0.75	0.43	0	1
Single-family house dummy	16,757	0.77	0.42	0	1

Notes: *Household Income: 01 = Less than \$15,000; 02 = \$15,000 to \$24,999; 03 = \$25,000 to \$34,999; 04 = \$35,000 to \$49,999; 05 = \$50,000–\$74,999; 06 = \$75,000–\$99,999; 07 = \$100,000–\$149,999; 08 = \$150,000 or more; Home Ownership: 0 = False (Rented); 01 = True (Owned).

² “The state of EVs in America”, Volvo Car USA.

Maryland's vehicles are powered by gasoline, with the remaining 6.3% distributed among other types. For the purposes of this study, we consider both battery electric vehicles (BEV) and plug-in hybrid electric vehicles (PHEV) as EVs. Out of the households surveyed, only 169 own at least one EV, and among these, 15 households possess two EVs. These EVs constitute 0.59% of the total vehicles in the sample data. Interestingly, households with EVs tend to own more vehicles, with an average of 1.85 vehicles for households without EVs, compared to 2.33 vehicles for households with EVs. Among the households with EVs, 81% of households also had a non-electric vehicle in addition to the EV (consistent with the findings from Davis (2023)).

Moreover, households with EVs tend to have higher household incomes on average. Additionally, households with EVs tend to be larger, with an average of 2.47 members compared to 2.19 members for non-EV households. These findings provide valuable insights into the characteristics and demographics of households that own EVs in the study sample.

3.2. Charging infrastructure

We utilized the electric vehicle charging station database provided by the U.S. Department of Energy (DoE) from the Alternative Fuels Data Center (AFDC). This publicly available national database contains comprehensive information about charging stations, including the number of charging ports based on charging speed, opening dates, geographic coordinates, and other relevant characteristics.

The database includes information about three types of charging ports: level-1, level-2, and DC fast chargers (DCFCs). Level-1 chargers are the slowest and, consequently, quite limited in public charging networks. On the other hand, level-2 chargers are more prevalent due to their higher power capacity and relatively affordable installation costs, making them a popular choice for public charging. DC fast chargers, being the fastest option as they employ high voltage, come with a higher usage cost. In Maryland, as of May 2019, there are a total of 1,203 public charging points available, with some stations having multiple charging outlets for different types of charging equipment. The distribution of these charging ports across the state includes 19 level-1, 967 level-2, and 217 DCFC ports.

Fig. 1 illustrates the spatial distribution of public charging ports installed in Maryland from 2010 to 2019. A noticeable concentration of chargers is evident in the Baltimore and Montgomery County (close to Washington D.C.) regions. It's important to note that these charging stations do not include residential, workplace, or private charging facilities. Additionally, we observe that more charging stations have been built in recent years.

We construct a measure of charging stations that calculates the total number of publicly accessible charging ports within a 5-mile radius from

the centroid of each census tract, assuming no heterogeneity across households regarding access to charging within the census tract. The choice of the 5-mile buffer is based on the 50th percentile of the daily average car trip distances. Furthermore, we recognize that the availability of level-2 and DCFC stations impacts EV adoption among consumers differently (Sommer and Vance, 2021). Level-2 charging points, due to longer charging times, are primarily utilized for opportunity charging or at locations like offices or homes for parking (Wolbertus and Van den Hoed, 2019). To address this in the model, we create a measure of level-2 charging by counting the number of such ports within a 1,000-meter radius from the centroid of each census tract.

On the other hand, despite having to travel a bit further, consumers prefer to use DC fast charging stations due to their relatively fast charging speed. To incorporate this charging preference heterogeneity, we create a measure of DC fast charging by counting the number of such ports within a 5-mile radius from the centroid of each census tract. This approach helps us capture the varying effects of charging station types on EV adoption in the study area.

Fig. 2 illustrates the 5-mile buffers created for two random census tracts. After constructing the data, on average, we find 45 level-2 charging ports and 4 DCFC ports within a 1,000-meter and 5-mile radius from the centroid of the census tract. Additionally, approximately 16% of the census tracts have no charging points available for public use.

4. Methodology

We develop an integrated approach to examine the relationship between the growth of EVs and the increased deployment of public charging stations. We start by estimating a binomial logit model

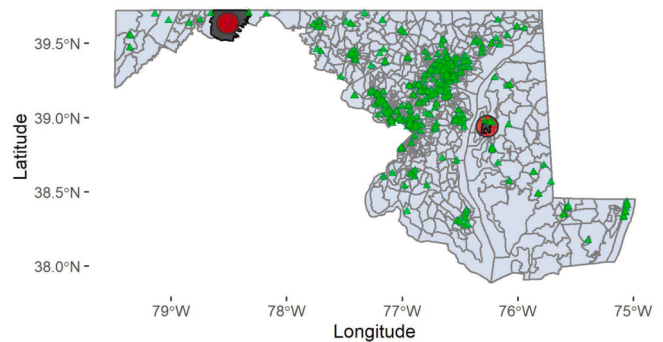


Fig. 2. Highlighting two census tracts with a 5-mile radius from the centroid. (The points in the map are the public charging station locations).

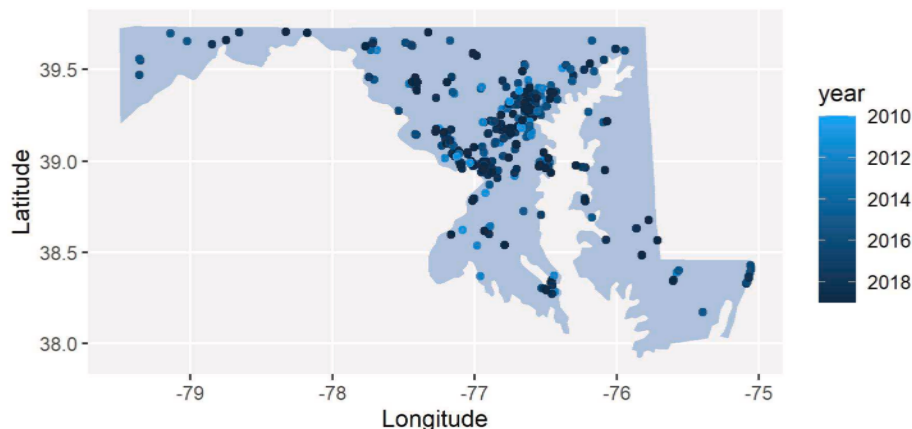


Fig. 1. Spatial distribution of public charging stations in Maryland.

utilizing household survey data to help us understand the factors influencing the decision to own an EV. These estimates are subsequently applied to a synthetic population generated using a Bayesian network-based method, enabling us to determine the probability of households owning an EV. Finally, we test various scenarios of growth in public charging stations on this synthetic population to observe the changes in EV ownership levels, taking into account housing types and location characteristics.

4.1. Binomial logit model

Drawing on the microeconomic theory of consumer behavior, the decision to own an EV can be framed as a household maximizing its utility, taking into account charging considerations and various other constraints. In this study, the choice set comprises two alternatives: owning an EV (both BEV and PHEVs), represented as $j = 1$, and not owning any EV, represented as $j = 0$. We denote the observed EV ownership of a household i during a specific period as Y_{ij} .

To model the household's decision regarding the ownership of an EV, we employ a binomial logit model (BNL) with a set of predictors. This modeling framework assumes that the utilities (U_j) associated with the alternatives in the discrete model consist of two components: a systematic utility component (V_j), which is observable, and an error term (ϵ_j), which is not observable. Following the discrete choice theory Train (2009), the probability that a household chooses EV ownership over not owning it, in terms of the binomial logit model, can be expressed by the following equation:

$$P_{i1} = \frac{1}{1 + \exp[V_{i1} - V_{i0}]} \quad (1)$$

Here, the utility of EV ownership can be written as $V_{i1} = \beta_0 + \beta_k X_k + \beta_c C$, where, X_k is a vector of consumer characteristics, location, and car characteristics, and C is the measure of public charging stations accessible to the consumer. β_k is a vector of parameters to be estimated and β_c reflects the role of public charging infrastructure. We assume households to be rational and choose to own an EV only if it maximizes utility.

The independent variables will include household income, the number of working persons in the household, whether they live in a single-family house, the number of bicycles owned by the household, the number of transit trips made during the travel day, the total distance traveled using a car during the travel day, whether they teleworked, and the possibility of charging at work. Additionally, the utility function of EVs will include a measure of level-2 and DCFC points that are available for public use in the corresponding census tract.

4.2. Synthetic population

Even though the survey dataset accurately represents the entire population of Maryland, it does not provide disaggregated data for every single unit in the population. For this research study, the weighted sample provided by the survey dataset was used to obtain the cumulative distribution functions (CDFs), which would later be utilized in constructing the synthetic population.

We employ a Bayesian network (BN) approach to create the synthetic population, an artificial representation of the actual population. It is a probabilistic model that encodes the joint probability distribution of a set of random variables and represents them in a directed acyclic graph (DAG) Koller and Friedman (2009). The DAG contains nodes and directional links that represent the variables and their conditional dependencies; the link $A \rightarrow B$ indicates that B (the child) is dependent on A (the parent). Each node carries the probability of finding that node under a given condition. Sun and Erath (2015) have used BN for creating a synthetic population for Singapore, and the study results showed that BN outperformed other population synthesizers such as Interactive Proportional Fitting (IPF) and Markov chain Monte Carlo algorithm

(MCMC) in terms of overfitting, heterogeneity, and missing data. Moreover, the BN approach has been used in recent studies to generate disaggregated data using traditional survey data or open data for transportation models (Zhang et al., 2019; Hörll and Balac, 2021).

A synthetic population was created, incorporating the necessary variables to accurately represent Maryland households in terms of income, household composition, travel behavior, and housing status. The sample observations for each census tract were converted to pseudo-observations ranging from 0 to 1 and used to learn the dependency structure among the selected variables of interest using Bayesian network learning techniques. To do so, we utilized a greedy search algorithm to determine the optimal network structure (see Fig. A1), establishing dependencies among the selected variables. The optimal network structure was then used to synthesize n number of synthetic pseudo-observations, where n is equivalent to the total number of households. Finally, the synthetic population was obtained by transforming the synthetic pseudo-observations to the original values using inverse CDFs. For more information on the BN method used in this paper to generate the synthetic population, see Jutras-Dubé et al. (2023).

As the sampling process may generate unfeasible samples, we incorporate a sanity check mechanism to exclude infeasible attribute combinations during the generation process. For example, we exclude any household sample if the number of workers exceeds the total number of people in that household or if the number of transit trips exceeds the total number of daily trips. By doing so, we ensure that the synthetic population remains realistic and representative of the underlying population. Our aim is to enhance the modeling process by synthesizing a population that encompasses a broader range of samples that might not be captured in the original survey data, thereby improving the robustness and generalizability of our models.

The Mean Absolute Percentage Error (MAPE) was used to compare the household attributes resulting from the synthetic population (synthesis estimates) with the weighted estimates from the RTS sample. MAPEs were found to be low for all variables, indicating a reliable synthetic population. Fig. 3 shows the results of a comparison between the synthetic population and the RTS sample.

5. Results

We start by estimating a binomial logit model that includes the total number of public charging ports accessible for households based on their location. Motivated by the differential effects of charging points by charging speed as discussed before, we subsequently analyze the effect of level-2 and DC fast charging. We then apply the model to the synthetic population generated to arrive at the probability of a household owning an EV.

Table 2 summarizes the final statistical results of the estimated binomial logit model. As mentioned, the estimates indicate the choice of households owning an EV. Model 1 presents the estimates which include a measure of total public charging ports accessible for the households, whereas Model 2 distinguishes between different levels of charging. Overall, the coefficients' signs were as expected and they were statistically significant in most cases.

A positive and significant coefficient of household income indicates that higher-income households are more likely to own an EV. Furthermore, a higher number of bicycles owned in the household has a positive influence on the ownership of an EV. The coefficient of the single-family house variable, which serves as a proxy for access to private charging at the home location, is positive but is statistically significant only at the 10% level. The coefficient of distance traveled using cars in the household is also significant, indicating that households with longer commutes are less likely to own an EV, reflecting concerns about the limited range of EVs.

Regarding charging infrastructure, the availability of public charging points is generally found to have a positive effect on EV ownership. In the model with only the total number of publicly accessible charging

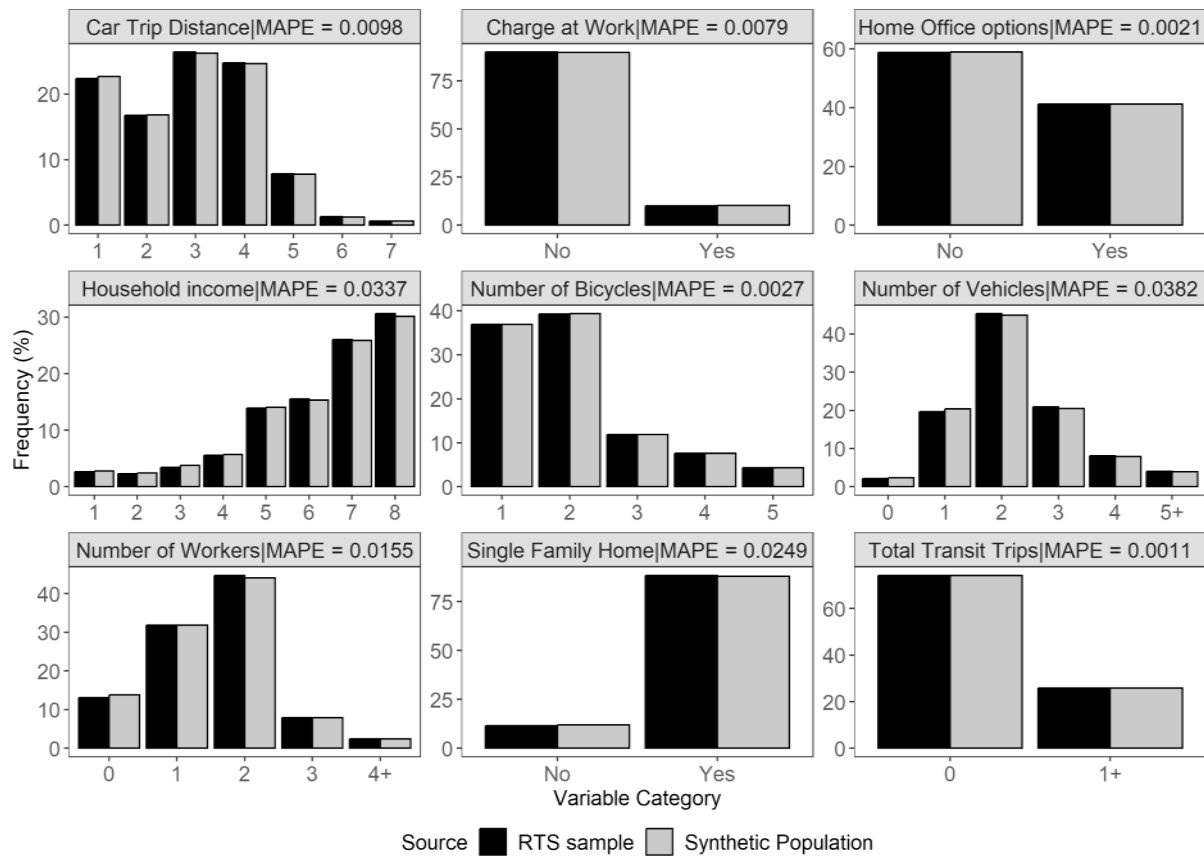


Fig. 3. Comparison of synthetic population and decennial census data.

Table 2
Estimates from the Binomial Logit model.

	Model 1		Model 2	
	Coeff.	Std. Err.	Coeff.	Std. Err.
Household income	0.303***	(0.065)	0.305***	(0.065)
Single-family house dummy	0.443	(0.290)	0.479	(0.294)
Number of bicycles	0.133**	(0.056)	0.136**	(0.056)
Number of workers	-0.264**	(0.107)	-0.269**	(0.107)
Total No. of transit trips	-0.034	(0.046)	-0.044	(0.046)
Total distance traveled by car	-0.005***	(0.002)	-0.004***	(0.002)
Home office dummy	0.789***	(0.179)	0.783***	(0.179)
Charge at work	0.795***	(0.197)	0.799***	(0.197)
No. of public charging ports	0.002	(0.001)	–	–
No. of level 2 charging ports	–	–	0.011	(0.014)
No. of DC fast charging ports	–	–	0.032***	(0.012)
Constant	-7.542***	(0.461)	-7.676***	(0.464)
No. of observations	30,868		30,868	
Initial log-likelihood	-1084.88		-1084.88	
Log-likelihood at convergence	-1026.83		-1023.78	
R-square	0.053		0.056	

Notes: Charging station variables in the model include: Level 2 charging ports within a 1,000-meter radius, DC fast charging ports within a 5-mile radius, and the total number of charging ports within a 5-mile radius from the centroid of the census tract. Additionally, the variable “charge at work” indicates the possibility of charging at the workplace for at least one person in the household. Household Income: 01 = Less than \$15,000; 02 = \$15,000 to \$24,999; 03 = \$25,000 to \$34,999; 04 = \$35,000 to \$49,999; 05 = \$50,000–\$74,999; 06 = \$75,000–\$99,999; 07 = \$100,000–\$149,999; 08 = \$150,000 or more; Home Ownership: 0 = False (Rented); 01 = True (Owned). Robust standard errors are reported in parenthesis. ***, **, *, and · denote statistical significance at the 0.1%, 1 %, 5 %, and 10 % level, respectively.

ports, the coefficient is positive but statistically insignificant. However, when differentiating between levels of charging stations, we observe that the availability of both level-2 and DC fast charging has a positive influence on EV ownership, while only the latter has a statistically significant influence. Specifically, the availability of DC fast charging has a stronger effect than level-2 charging (in line with the received literature Sommer and Vance (2021)), indicating households’ preference for accessibility to fast charging when owning an EV. A likelihood ratio test indicates that the specification with level-2 and DCFC variables (i.e., Model 2) in Table 2 provides a significantly better fit than the specification with omitting these variables.³

Interestingly, the possibility of teleworking and the possibility of charging at the workplace are both very significant and positively affect EV ownership. This suggests that households with access to teleworking options and workplace charging facilities are more likely to own an EV reflecting that flexibility in travel and convenience in access to charging is important. Notably, the most striking finding is that access to workplace charging has a stronger positive effect on EV ownership than publicly available DC fast charging, which is consistent with the findings from Sheldon et al. (2019) and Hardman et al. (2018).

Fig. 4 presents the marginal effects of key variables from Model 2 in Table 2. Having an additional DC fast charger within a 5-mile radius of the census tract centroid increases the probability of a household owning an EV by about 0.02%. Similarly, an additional Level-2 charging point within 1,000 meters increases the probability by 0.006%. Notably, access to workplace charging and the possibility of teleworking each independently increase the probability of EV ownership by 0.45%.

³ The likelihood-ratio (LR) test statistic equals 8.06 with a p-value of 0.01, indicating that the inclusion of level 2 and DCFC variables improves the model fit significantly.

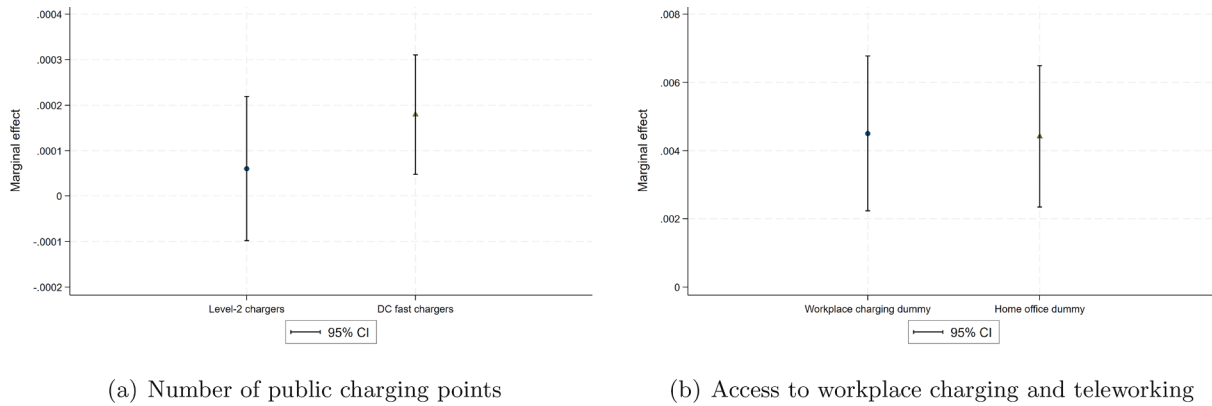


Fig. 4. Marginal effects of variables. *Notes:* The variables in Fig. 4a are the number of public charging points (No. of level-2 within 1000 meters and DCFC within a 5-mile radius from the centroid of the census tract) in our data, and the variables in Fig. 4b are dummies indicating the access to workplace charging and the possibility of teleworking.

Furthermore, to provide a better understanding of the practical significance of these findings, we follow the recommendation of Parady and Axhausen (2023) and calculate the marginal effects scaled by the standard deviation of the corresponding variables (standardized marginal effects). These estimates reveal that a one standard deviation increase in Level-2 chargers increases the probability of owning an EV by 0.002%, while a one standard deviation increase in DC fast chargers increases the probability by 0.003%. Access to workplace charging and home office possibility have a more substantial impact, with standardized effects of 1.7% and 0.9% increase in ownership probability, respectively.⁴

Due to the relatively low number of EVs in our survey data and the assumption of a symmetrical logistic distribution for the dependent variable in the logit model, we additionally estimated a skewed logistic model as a robustness check. While the point estimates from both models were similar, the standard errors differed slightly (see Table A1). This difference resulted in the effect of level-2 chargers becoming statistically significant in the skewed logistic model. The log-likelihood values between the two models were also quite comparable, as reported in the paper.

In the next step, using the estimates from Model 2, which distinguishes chargers by charging speed, we predict the probability of owning an EV for each household unit generated by the synthetic population. The predicted numbers of EVs are then aggregated for the entire state of Maryland and compared with the total number of EVs estimated from the survey to evaluate the accuracy of the results. The model predicts 15,809 EVs, compared to the 14,790 EVs reported in RTS, with a MAPE of 6.4%.

In Fig. 5, the spatial distribution of the predicted number of EVs across Maryland at the census tract level is illustrated. Overall, we identify a certain nexus between the distribution of charging stations (by levels) from Fig. 1, and the adoption of EVs among the synthetic population as shown in Fig. 5.

5.1. Scenarios of growth in public charging stations

Using the BNL model estimates and the synthetic population, we can evaluate different scenarios of charging station growth and assess their impact on EV ownership. Considering the emphasis of IIJA and IRA on deploying public charging stations in rural/disadvantaged communities and low-income areas, we predict the growth in EV ownership by a combination of increasing level-2 and DCFC stations across various areas

and income groups in Maryland. Regardless of the specific scenario, understanding how the EV market will respond to a certain increase in public charging stations is of significant interest to policymakers, transport planners, and automobile companies.

We start by classifying census tracts in Maryland into three area types: rural, suburban, and urban areas, based on household density (measured in households per square mile)⁵. Fig. 6 presents the predicted change in EV ownership across these areas in Maryland with a change in the current number of Level 2 and DCFC stations, assuming that all other household characteristics will remain the same. Overall, the results highlight two important findings: the growth in DC fast charging stations will significantly drive the increase in EV ownership across all area types, and for the same growth in charging stations, we observe a strong rise in the ownership of EVs in suburban areas.

As shown in Fig. 6, a 50-unit increase in Level 2 charging points across all census tracts, while maintaining the same level of DCFC points, results in an overall increase of 5,014 EVs in suburban areas, 4,608 EVs in urban areas, and only 820 EVs in rural areas. Similarly, a 50-unit increase in DCFC points across all census tracts leads to an overall increase of 26,735 EVs in suburban areas, 24,504 EVs in urban areas, and 4,406 EVs in rural areas. This indicates that for the same level of growth in charging stations, we observe more than a 5-fold increase in EV ownership levels across suburban and urban areas compared to rural areas.

The majority of the areas classified as urban are located in Baltimore and close to Washington D.C. (Montgomery and Prince George's County region) in Maryland, where reasonably significant public charging is available (see Fig. 1). As a result, we observe a significantly higher increase in EV ownership among households residing there, although it remains relatively lower compared to suburban areas.⁶ This difference in adoption rates could be attributed to the more extensive public transportation options available in these urban areas and the challenges of limited parking availability.

Regarding suburban areas, we find the highest increase in EV ownership levels with the growth in public charging stations, as accessible charging infrastructure can facilitate longer distance commutes for suburban residents. Conversely, due to longer commuting distances and

⁴ The mean and standard deviation of these key variables: level-2 chargers (0.496, 3.459), DC fast chargers (3.717, 5.958), workplace charging dummy (0.345, 0.475), and home office dummy (0.074, 0.261).

⁵ This classification follows the categories defined by Wieder (2019) as follows: rural area if the household density is less than 106, suburban area if the household density ranges from 106 to 1314, and urban area if the household density is greater than 1314. Based on this classification, the census tracts in Maryland are distributed as follows: 52% are categorized as urban areas, 37% as suburban areas, and 11% as rural areas.

⁶ Wood et al. (2023) also predicts stronger EV adoption levels in suburban levels across the US by 2030.

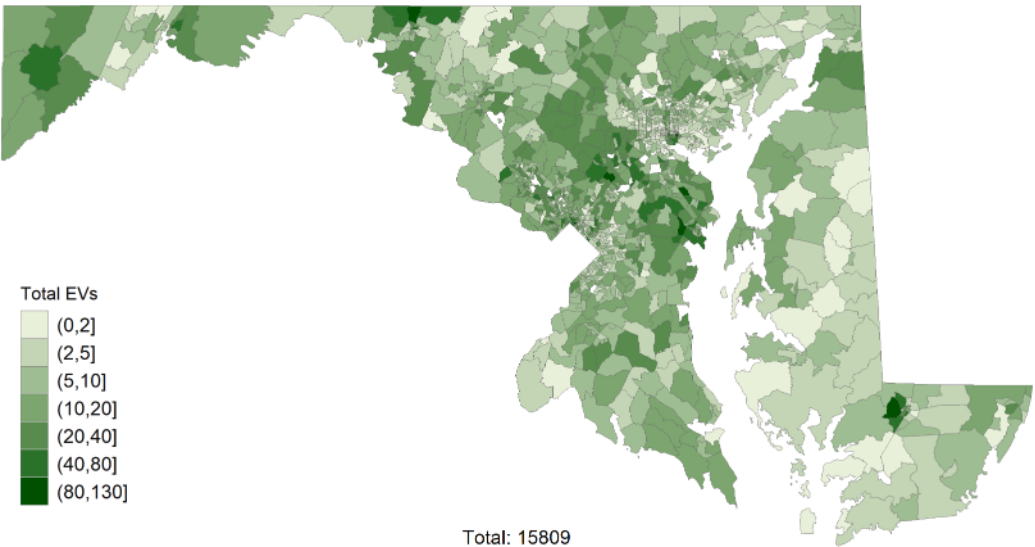


Fig. 5. Spatial distribution of the predicted number of EVs.

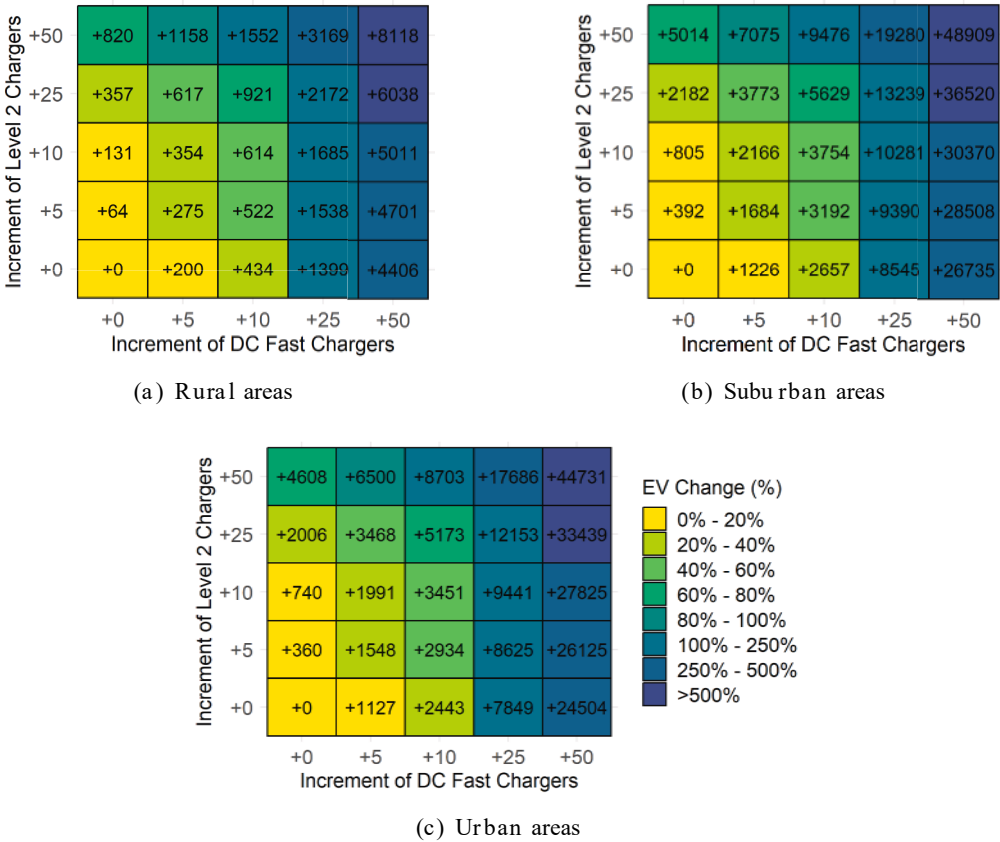


Fig. 6. EV ownership with an increase in public charging stations across area types.

sparingly distributed charging network, we observe relatively lower levels of EV ownership in rural areas. Consequently, the overall importance of growth in public charging points is higher in rural areas, followed by suburban areas, to achieve distributional equity.

Further, we categorize households into three groups based on their annual income, using criteria established by the Association for Neighborhood & Housing Development (ANHD) that relies on the Area Median Income (AMI). The categories are as follows: low-income households (income below 80% of AMI), moderate-income households

(income ranging from 80% to 160% of AMI), and high-income households (income at 160% of AMI or higher). For the year 2018 in Maryland, the AMI value was \$101,300, according to the Department of Housing and Urban Development (HUD).

Fig. 7 predicts the overall change in EV ownership with the growth in access to public charging stations across households of different income groups. Similar to the previous case, we observe that an equal unit increase in DCFC points has about five times more impact on EV ownership levels compared to Level 2 charging points across all income

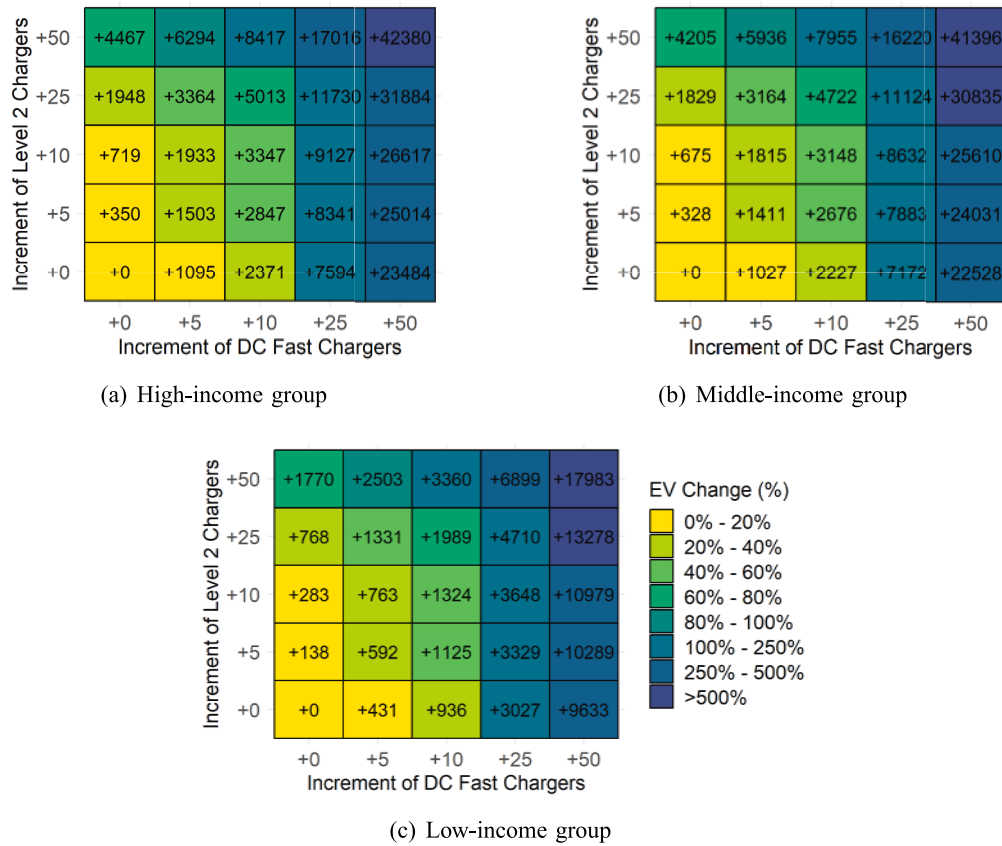


Fig. 7. EV ownership with an increase in public charging stations across income groups.

groups.

With an increase in access to public charging stations, high-income households show slightly higher adoption rates of EVs compared to middle-income households. Most of these high-income households are centered in/close to urban areas and can generally afford EVs, with access to charging facilitating long-distance travel and alleviating range anxiety.

As expected, low-income households exhibit a relatively lower increase in EV ownership with increasing access to public charging stations, mainly due to their higher upfront purchase costs. This suggests that improving access to public charging stations alone may not be sufficient to accelerate the ownership of EVs among low-income households. Additional financial incentives, along with access to infrastructure, are likely needed to encourage EV ownership among such households, thereby reducing the disparity in adoption of EVs.

6. Conclusion

Due to the limited availability of revealed preference survey data on electric vehicle choices, and the significance of understanding demand at a finer geographic level to better direct investments in public charging infrastructure, we propose an integrated approach of discrete-choice modeling and synthetic population generation through a Bayesian network-based approach. Using this method, we analyze the impact of public charging stations on electric vehicle ownership at the census tract level in Maryland. We differentiate between the effects of level-2 and fast chargers. Initially, we estimate a binomial logit model based on a regional travel survey to assess the likelihood of households owning an EV, while controlling for household demographics and nearby charging infrastructure.

Our findings align with existing literature, demonstrating a positive and significant effect of level-2 and DC fast chargers on EV ownership.

Notably, the impact of access to fast charging was stronger, reflecting its importance in facilitating longer-distance travel. Interestingly, access to workplace charging had a significantly positive effect, highlighting the crucial role of convenient charging options in influencing EV ownership. Additionally, the possibility of teleworking was found to be a positive factor for EV ownership, as it reduces the range anxiety for workers and provides for more flexibility in charging schedules.

Using our logit model estimates on the synthetic population-generated using a Bayesian network approach—we were able to predict EV ownership levels across various income groups and residential environments. Interestingly, for the same increase in public charging stations across census tracts, suburban areas exhibited the highest projected growth in EV ownership, followed by urban areas. Similarly, our model suggests a significant increase in EV adoption among high-income groups with more public charging stations, highlighting a potential disparity among households with different income levels.

By applying our model estimates to the state's synthetic population, we predicted a total of 15,809 EVs during the survey period. This approach crucially allowed us to capture the spatial distribution of EV adoption at the census tract level, thus providing insights into the geographically targeted developments in public charging facilities to promote equitable EV adoption.

We acknowledge that including interaction terms between public chargers and income/regional groups might reveal even more nuanced results. However, due to the limited sample size of EVs in our survey data (only 176 vehicles), including interactions would leave too little variation to estimate the parameters reliably. However, our synthetic population incorporates demographic information. This allowed us to predict the increase in EV ownership levels across income groups with an increase in public charging points. However, for a more detailed analysis of the differential effects across demographic groups, a larger real-world data sample would be beneficial for future research.

Although our methodology enables us to analyze EV ownership across the entire population, we acknowledge the reliance on slightly older data. The EV market and public charging stations have developed considerably since the data was collected, prompting the need for analysis using more recent data. Despite these limitations, our study provides valuable insights into the impact of public charging stations on EV ownership across different demographics.

As Maryland advances toward electrifying transportation, our findings will aid policymakers in deciding on charging infrastructure deployment and promoting equity in the EV market. Additionally, this methodology can be extended to other states or the United States if vehicle-type ownership models are available.

Appendix A

Table A1

Estimates from the Skewed Logistic (scobit) model

	Model 1		Model 2	
	Coeff.	Std. Err.	Coeff.	Std. Err.
Household income	0.303***	(0.073)	0.305***	(0.073)
Single-family house dummy	0.441	(0.291)	0.477	(0.296)
Number of bicycles	0.133**	(0.054)	0.135**	(0.054)
Number of workers	−0.263**	(0.114)	−0.268**	(0.114)
Total No. of transit trips	−0.034	(0.065)	−0.044	(0.065)
Total distance traveled by car	−0.005***	(0.002)	−0.004***	(0.002)
Home office dummy	0.786***	(0.176)	0.780***	(0.176)
Charge at work	0.791***	(0.194)	0.794***	(0.195)
No. of public charging ports	0.002	(0.001)	—	—
No. of level 2 charging ports	—	—	0.011*	(0.006)
No. of DC fast charging ports	—	—	0.031***	(0.010)
Constant	−16.715***	(0.828)	−16.923***	(1.996)
No. of observations	30,868		30,868	
Log-likelihood at convergence	−1026.78		−1023.76	

Notes: Charging station variables in the model include: Level 2 charging ports within a 1,000-meter radius, DC fast charging ports within a 5-mile radius, and the total number of charging ports within a 5-mile radius from the centroid of the census tract. Additionally, the variable “charge at work” indicates the possibility of charging at the workplace for at least one person in the household. Household Income: 01 = Less than \$15,000; 02 = \$15,000 to \$24,999; 03 = \$25,000 to \$34,999; 04 = \$35,000 to \$49,999; 05 = \$50,000–\$74,999; 06 = \$75,000–\$99,999; 07 = \$100,000–\$149,999; 08 = \$150,000 or more; Home Ownership: 0 = False (Rented); 01 = True (Owned). Robust standard errors are reported in parenthesis. ***, **, *, and · denote statistical significance at the 0.1%, 1 %, 5 %, and 10 % level, respectively.

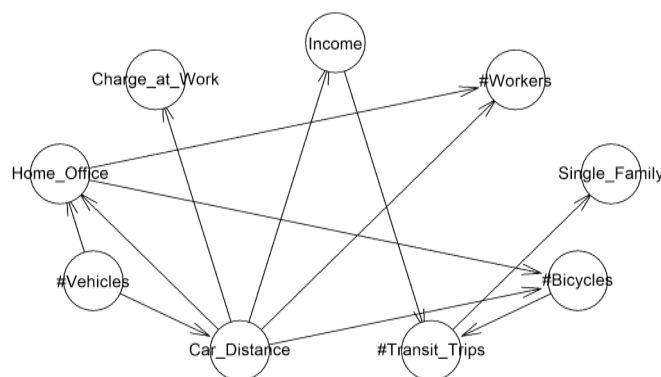


Fig. A1. Optimal Bayesian network structure reflecting the dependency among the variables.

References

- Barthelemy, J., Toint, P.L., 2015. A stochastic and flexible activity based model for large population application to Belgium.
- Bobeth, S., Matthies, E., 2018. New opportunities for electric car adoption: the case of range myths, new forms of subsidies, and social norms. *Effi.* 11, 1763–1782.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

We are grateful to Baltimore Metropolitan Council (BMC) for providing the Maryland travel survey data.

- Canepa, K., Hardman, S., Tal, G., 2019. An early look at plug-in electric vehicle adoption in disadvantaged communities in California. *Transp. Policy* 78, 19–30.
- Cole, C., Droste, M., Knittel, C.R., Li, S., Stock, J.H., 2021. Policies for electrifying the light-duty vehicle fleet in the United States. *JSTOR*.
- Davis, L.W., 2023. Electric vehicles in multi-vehicle households. *Appl. Econ. Lett.* 30 (14), 1909–1912.

- Egbue, O., Long, S., 2012. Barriers to widespread adoption of electric vehicles: an analysis of consumer attitudes and perceptions. *Energy Policy* 48, 717–729.
- Funke, S.A., Sprei, F., Gnann, T., Plötz, P., 2019. How much charging infrastructure do electric vehicles need? A review of the evidence and international comparison. *Transp. Res. Part D: Transport Environ.* 77, 224–242.
- Globisch, J., Plötz, P., Dütschke, E., Wietschel, M., 2019. Consumer preferences for public charging infrastructure for electric vehicles. *Transp. Policy* 81, 54–63.
- Greene, D.L., Kontou, E., Borlaug, B., Brooker, A., Muratori, M., 2020. Public charging infrastructure for plug-in electric vehicles: what is it worth? *Transp. Res. Part D: Transport Environ.* 78, 102182.
- Hackbarth, A., Madlener, R., 2013. Consumer preferences for alternative fuel vehicles: a discrete choice analysis. *Transp. Res. Part D: Transport Environ.* 25, 5–17.
- Hardman, S., Chandan, A., Tal, G., Turrentine, T., 2017. The effectiveness of financial purchase incentives for battery electric vehicles—a review of the evidence. *Renew. Sustain. Energy Rev.* 80, 1100–1111.
- Hardman, S., Jenn, A., Tal, G., Aksen, J., Beard, G., Daina, N., Figenbaum, E., Jakobsson, N., Jochem, P., Kinnear, N., et al., 2018. A review of consumer preferences of and interactions with electric vehicle charging infrastructure. *Transp. Res. Part D: Transport Environ.* 62, 508–523.
- Hörl, S., Balac, M., 2021. Synthetic population and travel demand for paris and ile-de-france based on open and publicly available data. *Transp. Res. Part C: Emerg. Technol.* 130, 103291.
- Hsu, C.-W., Fingerman, K., 2021. Public electric vehicle charger access disparities across race and income in california. *Transp. Policy* 100, 59–67.
- Jenn, A., Lee, J.H., Hardman, S., Tal, G., 2020. An in-depth examination of electric vehicle incentives: consumer heterogeneity and changing response over time. *Transp. Res. Part A: Policy Practice* 132, 97–109.
- Jutras-Dubé, P., Al-Khasawneh, M.B., Yang, Z., Bas, J., Bastin, F., Cirillo, C., 2023. Copula-based synthetic population generation. *arXiv preprint arXiv:2302.09193*.
- Koller, D., Friedman, N., 2009. *Probabilistic Graphical Models: Principles and Techniques*. MIT Press.
- Lee, J.H., Chakraborty, D., Hardman, S., Tal, G., 2019. Exploring heterogeneous electric vehicle charging behavior: mixed usage of charging infrastructure. Technical report.
- Levinson, R.S., West, T.H., 2018. Impact of public electric vehicle charging infrastructure. *Transp. Res. Part D: Transport Environ.* 64, 158–177.
- Li, L., Wang, Z., Chen, L., Wang, Z., 2020. Consumer preferences for battery electric vehicles: a choice experimental survey in china. *Transp. Res. Part D: Transport Environ.* 78, 102185.
- Li, S., Tong, L., Xing, J., Zhou, Y., 2017. The market for electric vehicles: indirect network effects and policy design. *J. Assoc. Environ. Resource Econ.* 4 (1), 89–133.
- Li, S., Zhu, X., Ma, Y., Zhang, F., Zhou, H., 2022. The role of government in the market for electric vehicles: evidence from china. *J. Policy Anal. Manage.* 41 (2), 450–485.
- Linn, J., 2022. Is there a trade-off between equity and effectiveness for electric vehicle subsidies. In: *Resources for the Future*.
- Lutsey, N., Slowik, P., Jin, L., 2016. Sustaining electric vehicle market growth in us cities. *Int. Council Clean Transp.*
- Narassimhan, E., Johnson, C., 2018. The role of demand-side incentives and charging infrastructure on plug-in electric vehicle adoption: analysis of us states. *Environ. Res. Lett.* 13 (7), 074032.
- Nazari, F., Mohammadian, A., Stephens, T., 2018. Dynamic household vehicle decision modeling considering plug-in electric vehicles. *Transp. Res. Rec.* 2672 (49), 91–100.
- Neaimeh, M., Salisbury, S.D., Hill, G.A., Blythe, P.T., Scoffield, D.R., Francfort, J.E., 2017. Analysing the usage and evidencing the importance of fast chargers for the adoption of battery electric vehicles. *Energy Policy* 108, 474–486.
- Nejad, M.M., Erdogan, S., Cirillo, C., 2021. A statistical approach to small area synthetic population generation as a basis for carless evacuation planning. *J. Transp. Geogr.* 90, 102902.
- Nicholas, M., Tal, G., 2017. Transitioning to longer range battery electric vehicles: implications for the market, travel and charging. *SAE Int.* 115, 1617–1622.
- Nicholas, M., Tal, G., Ji, W., 2017. Lessons from in-use fast charging data: why are drivers staying close to home. *Inst. Transp. Stud.*
- Parady, G., Axhausen, K.W., 2023. Size matters: the use and misuse of statistical significance in discrete choice models in the transportation academic literature. *Transportation* 1–33.
- Plötz, P., Funke, S.A., 2017. Mileage electrification potential of different electric vehicles in Germany. In: *European Battery, Hybrid and Fuel Cell Electric Vehicle Congress*, (pp. 1–8).
- Schulz, F., Rode, J., 2022. Public charging infrastructure and electric vehicles in norway. *Energy Policy* 160, 112660.
- Sheldon, T.L., 2022. Evaluating electric vehicle policy effectiveness and equity. *Annu. Rev. Resour. Econ.* 14, 669–688.
- Sheldon, T.L., DeShazo, J., Carson, R.T., 2019. Demand for green refueling infrastructure. *Environ. Resource Econ.* 74, 131–157.
- Sierzchula, W., Bakker, S., Maat, K., Van Wee, B., 2014. The influence of financial incentives and other socio-economic factors on electric vehicle adoption. *Energy Policy* 68, 183–194.
- Slowik, P., Lutsey, N., 2017. Expanding the electric vehicle market in us cities. *ICCT. Lett.* 16 (6), 064–092.
- Springel, K., 2021. Network externality and subsidy structure in two-sided markets: Evidence from electric vehicle incentives. *Am. Econ. J.: Econ. Policy* 13 (4), 393–432.
- Sun, L., Erath, A., 2015. A bayesian network approach for population synthesis. *Transp. Res. Part C: Emerg. Technol.* 61, 49–62.
- Train, K.E., 2009. *Discrete Choice Methods with Simulation*. Cambridge University Press.
- Wang, K., Zhang, W., Mortveit, H., Swarup, S., 2021. Improved travel demand modeling with synthetic populations. In: *Multi-Agent-Based Simulation XXI: 21st International Workshop, MABS 2020, Auckland, New Zealand, May 10, 2020, Revised Selected Papers 21*. Springer. pp. 94–105.
- Wieder, R.I., 2019. *Evaluating What Makes a us Community Urban, Suburban or Rural*. Decoded: Pew Research Center.
- Wolbertus, R., Van den Hoed, R., 2019. Electric vehicle fast charging needs in cities and along corridors. *World Electric Vehicle J.* 10 (2), 45.
- Wood, E., Borlaug, B., Moniot, M., Lee, D.-Y., Ge, Y., Yang, F., Liu, Z., 2023. The 2030 national charging network: estimating us light-duty demand for electric vehicle charging infrastructure. Technical report, National Renewable Energy Laboratory (NREL), Golden, CO (United States).
- Xing, J., Leard, B., Li, S., 2021. What does an electric vehicle replace? *J. Environ. Econ. Manage.* 107, 102432.
- Zhang, D., Cao, J., Feygin, S., Tang, D., Shen, Z.-J.M., Pozdnoukhov, A., 2019. Connected population synthesis for transportation simulation. *Transp. Res. Part C: Emerg. Technol.* 103, 1–16.